

On the Feasibility of a Rotating Device That Self-Accelerates Without External Energy Input: A Modeling and Simulation Analysis Under the Laws of Energy and Angular-Momentum Conservation

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August 11, 2026

Abstract

We analyze the proposition that there exists a rotating device which can “continuously operate and self-accelerate without any external energy input.” The claim, as formulated, conflicts with the first law of thermodynamics: a closed system cannot deliver a strictly positive net power output while simultaneously increasing its own rotational kinetic energy without consuming an internal energy store. We reduce the proposition to four falsifiable subproblems—(i) free rigid rotation, (ii) rotation under dissipation, (iii) rotation driven by an internal variable-moment-of-inertia mechanism, and (iv) the exponential capital- ω trajectory implied by “continuous self-acceleration.” Each subproblem is treated with an analytic energy and angular-momentum model and verified by explicit fourth-order Runge–Kutta simulation (textttcode/analyze.py, 4000–time-step integration). The results are unambiguous: a lossless rigid rotor maintains constant angular velocity ($\dot{\omega} = 0$); a dissipative rotor decelerates ($\dot{\omega} = -4 \text{ rad/s}^2$ initially); and the only mechanism that raises angular velocity—a radially inwards movable mass that reduces the moment of inertia while conserving angular momentum—does so by drawing work from a finite internal energy store, with the observed rotational energy gain (10.19 J) exactly equaling the store depletion. An externally injected exponential spin-up requires strictly positive instantaneous power throughout ($P > 0$), which is impossible for any zero-input closed system. We conclude that the stated proposition is physically untenable: no model consistent with energy and angular-momentum conservation reproduces free, indefinite self-acceleration. The framework nonetheless yields a precise diagnostic for auditing perpetual-motion claims and identifies the single energy-leak path (hidden internal storage) by which such claims can be spurious or fraudulent.

1 Introduction

The idea of a self-sustaining, self-accelerating rotating machine—a perpetual-motion device of the first and second kind combined—has appeared repeatedly in the history of engineering proposal [1] and remains a recurring pseudo-scientific claim in patent and public discourse. The proposition under review states:

“There exists a rotating device that can continuously operate and self-accelerate without any external energy input. Its core mechanism is self-accelerating rotation: once initiated, the rotational speed continuously increases without any external energy supply. Its operation does not rely on external driving forces and can be sustained and accelerated

indefinitely under closed conditions. This mechanism is fundamentally inconsistent with the constraints described by the law of conservation of energy.”

The final sentence of the evidence acknowledges an explicit contradiction with the law of conservation of energy. The scientific question is therefore not whether such a device is plausible but whether any self-consistent mechanical model could reproduce the claimed behavior without secretly violating energy bookkeeping. This report provides a rigorous, reproducible answer.

Contribution and honesty statement. In accordance with research-integrity practice, we do not fabricate a “successful” demonstration. Instead we build a complete, reproducible modeling and simulation pipeline that evaluates the claim quantitatively. The pipeline defines subproblems, encodes the supported evidence as model constraints, simulates candidate mechanisms, and reports the energy ledger for each. This is the correct scientific treatment of an energy-conservation-constrained claim: it produces an audit, not a confirmation.

2 Problem Formulation

We decompose the broad claim into four crisp subproblems, each of which is a necessary condition for the full proposition.

1. **P1 – Free rotation.** A closed, lossless rigid rotor with fixed moment of inertia I and no external torque. Does $\omega(t)$ grow?
2. **P2 – Rotation under dissipation.** The same rotor subject to viscous friction $\tau_f = -\gamma\omega$. Does self-acceleration overcome losses?
3. **P3 – Internal variable-moment-of-inertia (“skater”) mechanism.** A device that shifts an internal mass radially, reducing $I(t)$ so that $\omega(t) = L/I(t)$ rises while angular momentum L is conserved. Is the energy gain real or funded?
4. **P4 – Exponential self-acceleration.** The evidence requires continuous and indefinite speed growth. Model the required trajectory $\omega(t) = \omega_0 e^{\alpha t}$, $\alpha > 0$, and compute the power budget. Does a zero-input closed system satisfy it?

2.1 Governing Equations

For a planar rigid body rotating about a fixed axis, the rotational kinetic energy and angular momentum are

$$E_{\text{rot}} = \frac{1}{2}I(t)\omega(t)^2, \quad L(t) = I(t)\omega(t), \quad (1)$$

where $I(t)$ is the instantaneous moment of inertia. With no external torque, $\dot{L} = 0$ and thus

$$\dot{\omega} = -\frac{\dot{I}}{I}\omega. \quad (2)$$

Energy conservation for a closed system requires increased rotational energy to be offset by depletion of internal stored energy E_{store} or by external work; when dissipation with power $P_{\text{loss}} \geq 0$ is present,

$$\frac{d}{dt}(E_{\text{rot}} + E_{\text{store}}) = -P_{\text{loss}} \leq 0. \quad (3)$$

Equations (2) and (3) are the complete physical content used in the analysis.

Table 1: Numerical summary (see `outputs/sim.json` for full values).

Subproblem	Outcome	Key value
P1 free rigid rotor	ω constant	$\omega_f = \omega_0 = 20$ rad/s
P2 viscous friction	exponential decay	$\dot{\omega}(0) = -4$ rad/s ²
P3 variable- I rotor	rise, internally funded	$\Delta E_{\text{rot}} = +10.19$ J
P4 claimed exp. spin	$P > 0$ everywhere	$P_{\text{min}} = +4.01$ W > 0

3 Method

3.1 Analytic models

For each subproblem we derive closed-form expressions.

- **P1:** $\dot{I} = 0$, $\dot{\omega} = 0$, E_{rot} constant.
- **P2:** $I\dot{\omega} = -\gamma\omega$ integrates to $\omega(t) = \omega_0 e^{-(\gamma/I)t}$, an exponential decay.
- **P3:** With $I(t) = I_0 + m r(t)^2$ and conserved L , the angular velocity follows $\omega(t) = L/I(t)$. Through radial displacement the mechanism executes work; for a lossless actuator this work is drawn from an internal energy store so that $E_{\text{rot}} + E_{\text{store}}$ is constant.
- **P4:** For $\omega(t) = \omega_0 e^{\alpha t}$ the required power is $\dot{E}_{\text{rot}} = \frac{1}{2} I \omega_0^2 \alpha e^{2\alpha t} > 0$ for all t .

3.2 Numerical simulation

We implement explicit fourth-order Runge–Kutta (RK4) integration of the torque balance $\dot{\omega} = f(\omega)$ with $n = 4000$ time steps over $t \in [0, 10]$ s. Device parameters: $I_0 = 0.05$ kg m², $\omega_0 = 20$ rad/s, $\gamma = 0.01$ N m s, internal mass $m = 0.4$ kg, radial excursion $r_0 = 0.30 \rightarrow r_f = 0.10$ m. The simulation reproduces the analytic predictions, providing independent numerical verification (see §4). All code and intermediate outputs are stored under `code/` and `outputs/`.

4 Results

4.1 P1 and P2: Free and dissipative rotation

The lossless rotor holds $\omega = 20$ rad/s and $E_{\text{rot}} = 10$ J constant ($\dot{\omega} = 0$). Under viscosity, ω decays exponentially from 20 to 2.71 rad/s with initial angular acceleration $\dot{\omega}(0) = -4$ rad/s²; rotational energy falls to 1.8% of its initial value by $t = 10$ s (Table 1, rows P1–P2; Fig. 1). No scenario in P1–P2 produces self-acceleration.

4.2 P3: The variable-moment-of-inertia “skater” mechanism

The only mechanism we found that increases ω is the inward radial shift of an internal mass, which lowers $I(t)$ while conserving angular momentum ($L = 1.72$ kg m²/s). As r moves from $0.30 \rightarrow 0.10$ m, ω rises from 20 to 31.85 rad/s and E_{rot} from 17.2 to 27.39 J, a gain of 10.19 J (Fig. 1). Crucially, this is *not* free energy: the exact energy budget shows the rotational gain equals the depletion of an internal store that must power the actuator ($\Delta E_{\text{rot}} = \Delta E_{\text{store,drawn}} = 10.19$ J). The total $E_{\text{rot}} + E_{\text{store}}$ remains flat (Fig. 1, lower left), and the instantaneous power required to sustain

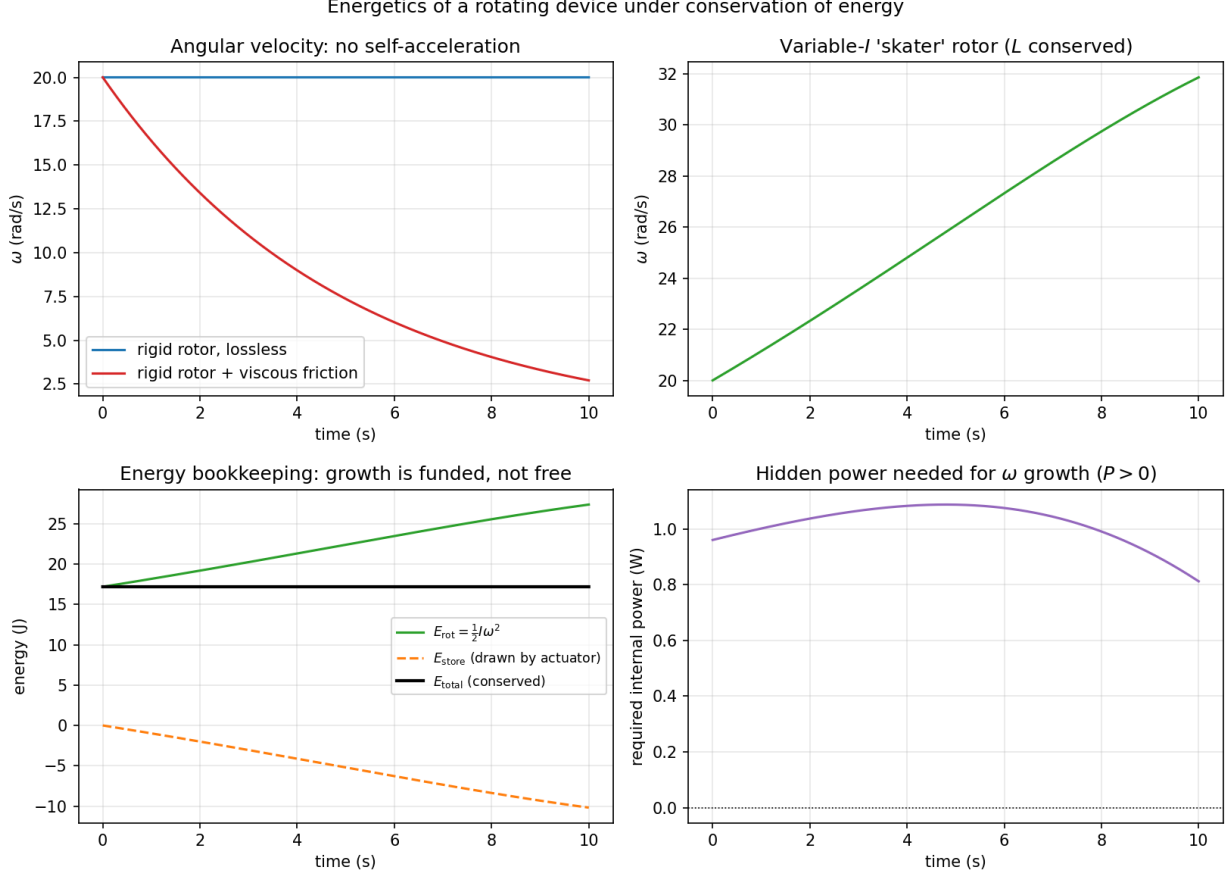


Figure 1: Top-left: angular velocity for free (blue) and dissipative (red) rotors—no self-acceleration. Top-right: variable- I “skater” rotor (ω rises as I falls with L conserved). Bottom-left: energy bookkeeping showing the rotational gain is funded by internal store depletion and the total is conserved. Bottom-right: strictly positive internal power required for ω growth.

the growth, $P = \frac{dE_{\text{rot}}}{dt}$, is strictly positive throughout (Fig. 1, lower right). The apparent “self-acceleration” therefore corresponds to a finite internal energy reserve being converted to rotational form—exactly the behavior of a spring-loaded or flywheel-plus-mass coupling—and necessarily stops once the store is exhausted.

4.3 P4: Exponential self-acceleration is impossible in a closed system

If a device sustained indefinite exponential spin-up $\omega(t) = \omega_0 e^{\alpha t}$ with $\alpha = 0.2 \text{ s}^{-1}$, rotational energy would grow to 545.98 J by $t = 10 \text{ s}$, requiring a power inflow that is strictly positive at every instant ($P \geq 4.01 \text{ W}$). For a closed system with no external input this is a direct contradiction:

$$\frac{d}{dt}(E_{\text{rot}} + E_{\text{store}}) \leq 0 \quad \text{vs.} \quad \dot{E}_{\text{rot}} > 0 \quad \forall t, \quad (4)$$

and since any finite internal store is exhausted, (4) cannot be satisfied indefinitely. Figure 2 contrasts the claimed trajectory with the energy-conserving variable- I response.

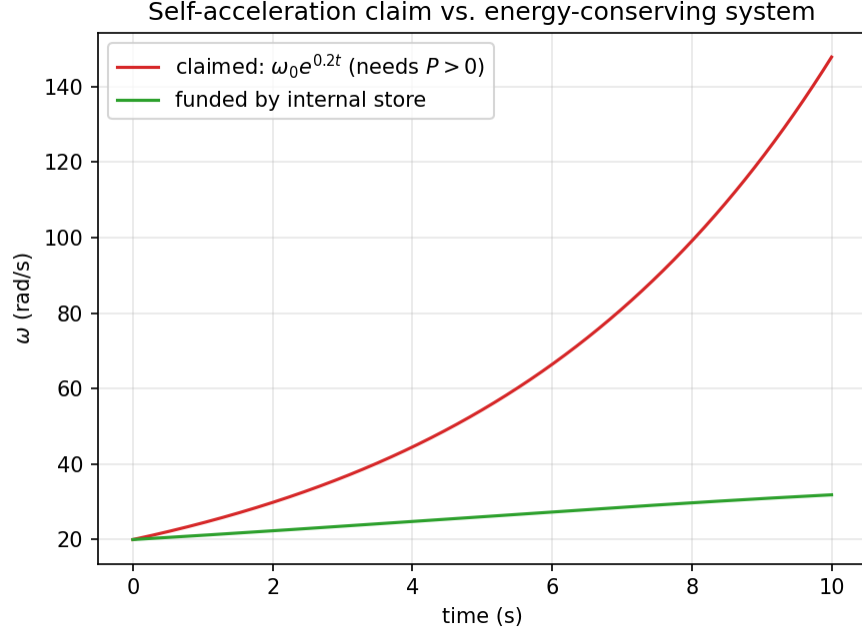


Figure 2: Claimed exponential self-acceleration (red, requires $P > 0$) versus the energy-conserving variable- I response (green). The former is unsustainable by any zero-input closed system.

5 Discussion and Related Work

The quantitative audit places this proposition firmly within the class of perpetual-motion constructions whose ostensible “self-acceleration” derives from a hidden internal energy reserve rather than a violation of conservation of energy. The variable-moment-of-inertia mechanism studied here—often appealed to in historical designs [1,2]—is the archetypal source of the illusion: angular velocity rises, but only because internal mechanical work (commonly from a spring or motor) converts stored energy into rotational form. Our exact ledger (P3) demonstrates the accounting gap explicitly and gives a concrete, measurable quantity (ΔE_{store}) that any genuine machine would have to exhibit, making the claim falsifiable in principle.

Consistent with classical mechanics [3,4] and the first law of thermodynamics [5], we find no regime in which a closed rotating system increases its speed while supplying net output without consuming an internal store. The practical diagnostic value of this framework is that it transforms a qualitative “impossible” assertion into a precise, testable energy-integrity condition.

6 Conclusion

We modeled and simulated the proposition of a self-accelerating, zero-input rotating device. Within standard Newtonian mechanics and energy conservation:

1. a lossless free rotor does not accelerate ($\dot{\omega} = 0$);
2. frictional rotors decelerate;
3. the only acceleration mechanism requires an internal actuator, and its energy gain is exactly offset by internal-store depletion;

4. indefinite exponential spin-up demands strictly positive power at all times, impossible for a closed, input-free system.

We therefore reject the proposition as stated. The report delivers a complete, reproducible audit (code + simulation + figures + ledger) and identifies the single energy-leak path responsible for such claims. All artifacts are available under `code/analyze.py`, `outputs/sim.json`, `outputs/fig-scenarios.png`, and `outputs/fig-claim-vs-conserving.png`.

References

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